



DPD zero

SQL MURDER MYSTERY

A Data-Driven Investigation

TechNova Inc. | CEO Office | October 15, 2025 | 21:00 Hrs

Investigation Date: October 15, 2025 | Author: Abhishek Pawar | Investigation Database: murder_mystery



STEP 1: Crime Scene Analysis

Objective: Identify presence in CEO Office during crime window (20:30–21:30)

Key Findings:

- ✓ Only **ONE employee** detected in CEO Office
- ✓ Entry time: **20:50** | Exit time: **21:00**
- ✓ Employee ID: **4 - David Kumar**

```
SELECT * FROM keycard_logs WHERE room = 'CEO  
Office'  
  
AND entry_time ≤ '2025-10-15 21:30:00'  
  
AND exit_time ≥ '2025-10-15 20:30:00';
```



STEP 2: Critical Area Access

Objective: Verify who accessed CEO Office & Server Room at time of crime

Evidence:

- ✓ **Exclusive Access:** David Kumar was the ONLY person in CEO Office (20:50–21:00)
- ✓ No other employees crossed these critical zones during the time window
- ✓ **Damning Detail:** Timing matches reported time of death

```
SELECT e.name, k.room, k.entry_time FROM keycard_logs k
JOIN employees e ON k.employee_id = e.employee_id
WHERE '2025-10-15 21:00:00' BETWEEN k.entry_time AND
k.exit_time;
```



STEP 3: Alibi Verification

Objective: Compare claimed locations vs actual keycard logs

The Contradiction:

David Kumar's Claim: "I was in the Server Room"

ACTUAL LOCATION: CEO Office (20:50–21:00)

→ **ALIBI IS FALSE** ✗

```
WITH claimed AS (...) SELECT c.employee_id,  
c.claimed_location,  
a.room AS actual_location FROM claimed c JOIN actual a...
```



STEP 4: Call Activity Analysis

Objective: Identify suspicious calls during crime window

Incriminating Call:

Caller: David Kumar

Receiver: Alice Johnson

Time: ~21:00 (during crime window) | **Duration:** 45 seconds

⚠ **Call made WHILE in CEO Office**

```
SELECT c.*, caller.name, receiver.name FROM calls
c
WHERE c.call_time BETWEEN '2025-10-15 20:30:00'
AND '2025-10-15 21:30:00';
```



STEP 5: Physical Evidence

Objective: Connect evidence with suspect movements

Evidence Trail Links to David Kumar:

- ✓ **Evidence #1–2:** Fingerprints & traces in CEO Office (20:50–21:00)
- ✓ **Evidence #3:** Digital traces match Server Room access (earlier that day)
- ✓ **Timeline Match:** All evidence places him at crime scenes at critical times

```
SELECT ev.*, k.employee_id, e.name FROM evidence ev
LEFT JOIN keycard_logs k ON ev.room = k.room
WHERE ev.found_time BETWEEN '2025-10-15
20:50:00'...
```

INVESTIGATION COMPLETE



VERDICT

DAVID KUMAR
IS THE CULPRIT

The SQL Query That Solved It:

```
WITH present AS (SELECT employee_id FROM keycard_logs WHERE room = 'CEO Office'...),  
    lied AS (SELECT employee_id FROM alibis WHERE claimed_location != 'CEO Office'),  
    calls_nearby AS (SELECT caller_id FROM calls WHERE call_time BETWEEN '20:50' AND '21:05')  
    SELECT emp.name FROM suspects s JOIN employees emp...
```

- ✓ **Presence:** CEO Office during crime window
- ✓ **Motive:** False alibi indicates premeditation
- ✓ **Evidence:** Physical traces + Call activity + Location data

Investigation Closed | Database: murder_mystery | Analysis Complete